Long Data

Improving access to and use of the city's archival records

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Long data

Big data means many records but typically over a short span

Municipal records often run continuously for decades → *long data*

Long data can illustrate trends, establish precedents, enable comparisons, and uncover context

Long data also poses challenges around accessibility, standards, comparability

Holding	Accessible?	Amount
Digital files with text	Yes	Small
Scanned pages	No	Large
Paper, film, audio	No	Unknown

Samuel Arbesman, "Stop Hyping Big Data and Start Paying Attention to 'Long Data'," *Wired*, January 29, 2013

Document understanding

Document understanding is a research field and software frameworks that turn documents into data

- **Layout analysis** finds the headings, paragraphs, tables, and figures on a page, and the order to read them in
- **Optical character recognition (OCR)** reads printed or handwritten text from a picture of a page
- **Table recognition** rebuilds a table's rows, columns, and cells
- **Field extraction** pulls named items like dates or totals out of forms

Contemporary AI models have rapidly improving ability to convert complex PDFs into more accessible text, markdown, HTML formats



From `https://www.datalab.to`

Data liberation and rescue

Data liberation moves information that is already public out of the formats that bury it (scans, PDFs, portals) and into open and accessible data formats

Data rescue is a grassroots interdisciplinary movement to preserve public datasets from decay and censorship

- **2014.** The Sunlight Foundation runs a PDF Liberation Hackathon to build shared tools for pulling tables out of government PDFs
- **2017 on.** The Public Utility Data Liberation project turns federal energy filings into an open and documented database
- **2020.** SCOTUS rules in *Georgia v. Public.Resource.Org* that the state's annotated code cannot be copyrighted
- **2025 on.** Researchers improvise technical capacity to archive data before it is deleted by the Trump administration

Tom Lee, "PDF Liberation: Why It Matters and How You Can Help," Sunlight Foundation, January 24, 2014; *Georgia v. Public.Resource.Org, Inc.*, 590 U.S. 255 (2020)

Data liberation for the City of Boulder

- **Hotline answers.** Aiding staff in research to answer questions from council and other departments
- **Budget history.** Converting PDF tables into accessible data formats
- **The law on a date.** Which version of an ordinance was in force when a permit was issued
- **Records requests.** A document made searchable once is ready for every later request that asks for it
- **Accessibility.** Every scanned page gains text that a screen reader can speak
- **Research and reporting.** Residents, reporters, and the university work from the same corpus as staff

Two data liberation sprints

I have already piloted several data liberation sprints using local government archives

City budget books, 2002–2027

City budgets

34 PDFs and 10,843 pages from city records.

6 volumes needed to be OCR'd

450 values traced to source documents and reconciled

CDE enrollment yearbooks, 1986–2025

Pupil membership

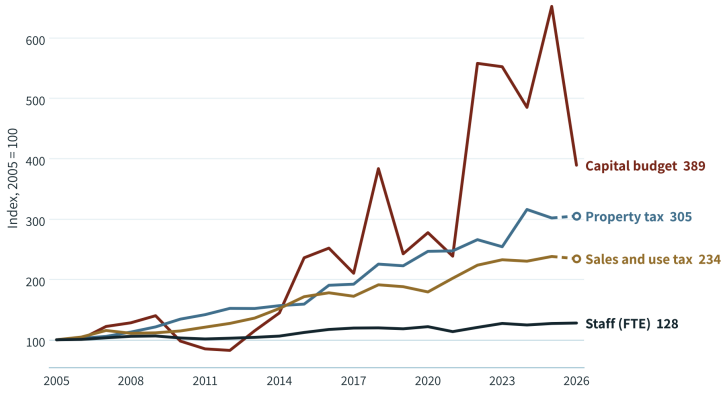
Scanned tables of enrollment by grade and school district

OCR and table extraction using Datalab

1,269 values extracted and audited for BVSD and SVVSD

Agent pipelines to identify page ranges, orchestrate OCR, extract tables, audit margins, record gaps, compare across volumes

Twenty-two years of the city's budget



Indexed to 2005 = 100. Taxes: audited collections through 2025, 2026 adopted (dashed). Capital: adopted capital budget. Staff: adopted FTE. Nominal dollars. City of Boulder budget books and annual financial reports, via budget-history dataset

Teaching with city's records portal

Week 09 of my Web Data Science course uses Finance Department's year-end revenue reports:
`https://cuinfoscience.github.io/Web-Data-Science-Book/ch-09-pdfs.html`

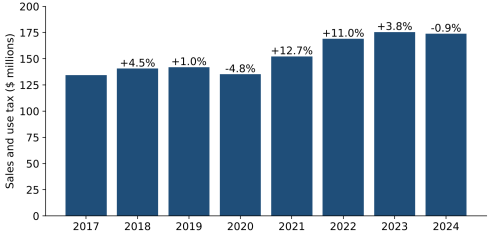
Access city's Laserfiche archives, download report PDFs, find and extract tables, check arithmetic against reported margins and prior years

Table 1 - Sales and Use Tax

All Funds

Sources - Related to Current Economic Activity	YTD through December			
	2017	2018	$ Change	% Change
Sales Tax	$104,144,137	$106,614,788	$2,470,651	2.37%
Rec MJ Additional 3.5%*	1,324,935	1,775,066	450,131	33.97%
Business Use Tax	14,216,145	13,070,617	(1,145,528)	(8.06%)
Construction Use Tax	9,897,738	13,018,049	3,120,311	31.53%
Motor Vehicle Use Tax	4,496,393	4,853,318	356,925	7.94%
Total Sales & Use Tax based on current economic activity	**$134,079,348**	**$139,331,838**	**$5,252,490**	**3.92%**
Sources - Related to Prior Periods				
Sales Tax Audit Revenue	189,929	211,748	21,819	11.49%
Business Use Tax Audit Revenue	269,615	1,105,124	835,509	309.89%
Total Sales and Use Tax	**$134,538,892**	**$140,648,710**	**$6,109,818**	**4.54%**

* Additional sales tax on Recreational Marijuana is dedicated to the general fund



Council hotline

Council members and staff exchange questions and answers via council hotline

797 messages since March 2023: 209 questions from council and 144 answers from staff

Many of those answers point to budgets, presentations, ordinance, and reports already in the city's archives

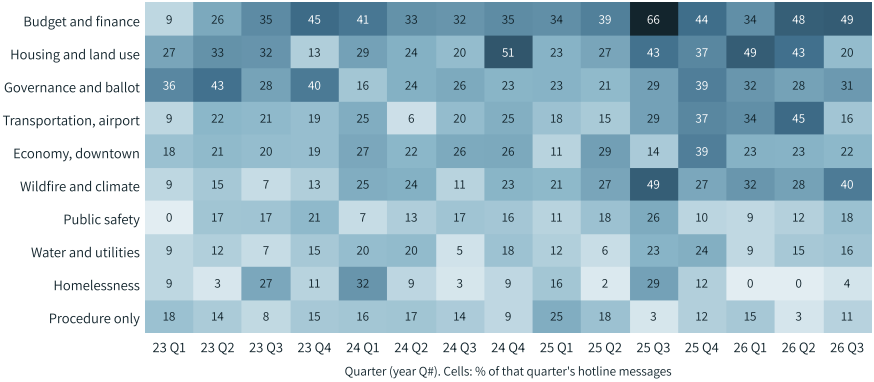
Question

How many of council's questions could be answered with staff-supervised automation using the city's own records?

City of Boulder open data, "Emails to the City Council Hotline," March 2, 2023 to October 15, 2026

Analyzing hotline: questions from council

Classify each message's subject and body based on keywords into nine policy themes



Code, keyword lists, and labels: `code/long-data-analysis.ipynb`, `code/hotline-labels-all.csv`. Messages through October 15, 2026; the chart shows complete quarters

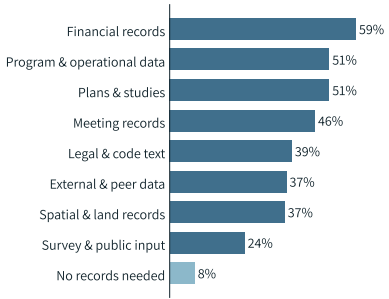
Analyzing hotline: answers from staff

Classify each staff answer's subject and body based on keywords into eight kinds of records it draws on

Of 144 answers, the most common: plans and studies (47%), financial records (43%), program data (39%), legal and code text (38%)

56% draw on three or more kinds of records

Council's questions mention financial records more often (70%) than staff answers do (43%)



What to liberate first

How often council needs the record? How far back it runs? How much personal information?

Start now

- Budgets, revenue, audits
- Council packets
- Staff memos
- Ordinances, resolutions
- Code versions
- Reports and plans

Next

- Minutes and recordings
- Board and commissions
- Program year-end reports

Only with screening

- Permits and land records
- Contracts
- Emails

Accessibility and CORA/FOIA

ADA requires the city's web content, including PDFs, to meet WCAG 2.1 in 2027

Scanned documents without text layers and figures without alt-text are endemic throughout municipal archives

Documents needed for service must be readable while archived scans are exempt

Fulfilling CORA requests can be expensive and complex → investments in data liberation and rescue could be a longer-term cost saving center



28 C.F.R. §§35.200–35.201, as amended by 91 Fed. Reg. 20902 (April 20, 2026); C.R.S. §24-34-802 (HB21-1110, HB24-1454)

Community values

The city's Sustainability, Equity and Resilience Framework commits the city to a government that:

- "Models stewardship, resilience, and sustainability of the city's financial, human, **information**, and physical assets"
- "Supports strategic decision-making with … timely and reliable data and analysis"
- "Enhances and facilitates **transparency**, accuracy, efficiency, effectiveness... in all city business"
- "Disaggregates data by race and ethnicity to ensure programs, policies, and financial decisions do not have a disparate impact on historically excluded community members"

Information like records are a city assets, alongside money, people, and buildings

Uneven digitization and accessibility limits ability for electeds, staff, researchers, journalists, *etc.* to study disparate impacts

City of Boulder, "Sustainability, Equity and Resilience Framework."

What happens next

Reading a scanned page costs under a cent today, and the price keeps falling

Finding the right document, checking what the machine read, and deciding what may be released will still take people. Retention schedules and legal review don't get cheaper with faster chips

Vendors will arrive with products running on the city's own documents. For the hotline, the product worth buying writes a draft answer with citations, which a staff member checks and signs



Buying a chatbot to answer council's questions

Publishing the records its answers would cite

Why I might be wrong

Demand may be low. My evidence is one coder and 40 hotline answers

Models may outgrow search. If a model can read every record at once, the city needs less search machinery, though it still needs every answer to cite a page

An automated assistant that chains several lookups can go wrong at each step without flagging it. Nobody has measured how often on city records

Most likely of all, the project just stops: multi-year records work dies on one budget cycle or one staff departure



Some records shouldn’t become data

Sort every holding into one of three kinds: **already data**, **could become data**, or **shouldn’t become data**

The third kind includes surveillance video, utility and recreation customer records, code complaints that name residents, protective-order addresses, and records about victims and juveniles

The danger is aggregation. Each record can stay obtainable on its own under CORA, while a searchable index would let anyone assemble a dossier in minutes

The city has done this before: the published hotline replaces every email address and phone number with a one-way code



“With great power comes great responsibility”

Who bears the cost

What gets scanned gets cited. Records left on paper drop out of every search and analysis, and the least-digitized records tend to describe the least powerful residents

OCR does worst on old, handwritten, tabular, and non-English pages, so search quality drops along the same lines

A searchable index exposes residents with the most contact with code enforcement and police far more than anyone else



Set the order on purpose

Let equity decide which holdings are digitized first. The city's own framework already asks it to check data for disparate impact

Key takeaways

1. The city holds **long data**, and most of it a computer can't yet read
2. Reading pages is cheap now; **finding and checking** them is the work
3. The **hotline** shows the demand: a third of substantive answers were lookups in records the city already holds
4. **Accessibility law, CORA, and the city's own framework** already call for this work
5. Before buying anything: **an inventory**, stable document IDs and bulk export written into contracts, and a rule that every machine answer cites a page

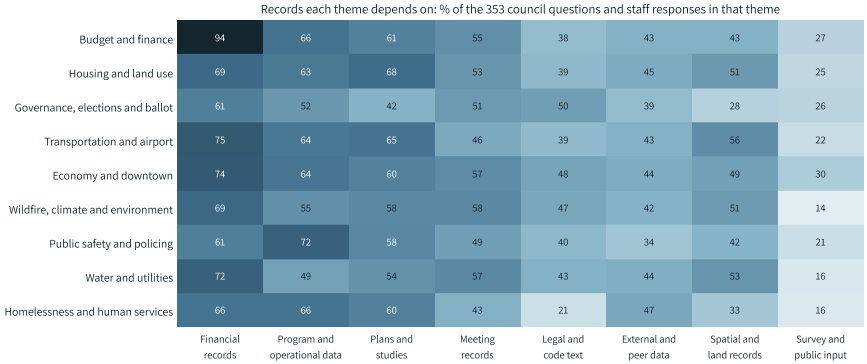
A test for the first demonstration

Which of the three kinds of holding is its corpus? Does every answer cite a document and page?

Appendix: record priorities in detail

Record	Serves	What we know of its state	Tier
Budget books	Financial questions; 26-year series	2002–2027 in the records portal; 1,651 pages lack text	Start
Revenue reports	Financial questions	In the records portal, 2018–2024; also sent as hotline attachments, which are not in the dataset	Start
Council packets and memos	Meeting records; plans; the answers themselves	Recent years in the agenda system; depth unknown	Start
Ordinances, resolutions, code versions	Legal text; which law applied when	Current code hosted outside the city; history in the records portal	Start
Minutes and recordings	Deliberation and intent	Unknown; needs inventory	Next
Plans and consultant studies	Plans and studies	Comprehensive plan: 15 editions converted, 1977–2026	Next
Board and commission appointments	Who served on what, when	Announced on the hotline; rosters unknown	Next
Program year-end reports	Operational data	Some sent as hotline attachments	Next
Permits and land records	Where things are	Department systems; custody split	Screen
Hotline and public comment	What council asks; deliberation	904 records, March 2023 to October 15, 2026; email addresses and phone numbers coded	Screen

Appendix: which records each theme depends on



Financial records come up in more than 60% of questions and answers in every theme. Most answers need several kinds of record at once

Appendix: getting a report from the records portal

```
import requests, time
# PORTAL, LISTING, and DOWNLOAD hold the portal's addresses; chapter 9 gives them in full
HEADERS = {"User-Agent": "WebDataScience/1.0 (your-email@colorado.edu)"}
# The folder's listing: the request the folder page itself sends (194345 is Revenue Reports)
body = {"folderId": 194345, "getNewListing": True, "start": 0, "end": 100}
response = requests.post(PORTAL + LISTING, json=body, headers=HEADERS, timeout=60, verify=bundle)
for entry in response.json()["data"]["results"]:
   print(entry["entryId"], entry["name"])

time.sleep(30) # wait between requests to the city's server
response = requests.get(PORTAL + DOWNLOAD.format(docid=194397), headers=HEADERS, timeout=120, verify=bundle)
print(response.headers["Content-Type"], len(response.content), response.content[:5])
# application/pdf 723920 b'%PDF-'
```

`bundle` is a certificate file the book provides, because the city's server is set up in a way browsers forgive and Python doesn't. Chapter 9 also sends the request fields the page adds

Appendix: reading and checking the totals row

```
import pdfplumber

def row_words(page, label):
   """Return the words to the right of label, on the line where label is."""
   found = page.search(label)[0] # where the label sits on the page, in points
   line = page.crop((0, found["top"] - 1, page.width, found["bottom"] + 1))
   return [w["text"] for w in line.extract_words() if w["x0"] > found["x1"]]

for name in REPORTS:
   with pdfplumber.open(name) as pdf:
      page = find_summary_page(pdf) # the page whose text includes "This report provides"
      earlier, this, change = [to_number(t) for t in row_words(page, "Total Sales and Use Tax")[:3]]
   if this - earlier != change: # the report's own change column is the check
      print(f"{name}: {this:,} - {earlier:,} is not {change:,}")
# (nothing printed: every change column checks)
```

`to_number` turns "$134,538,892" or "(1,145,528)" into a whole number; accountants write a negative number in parentheses